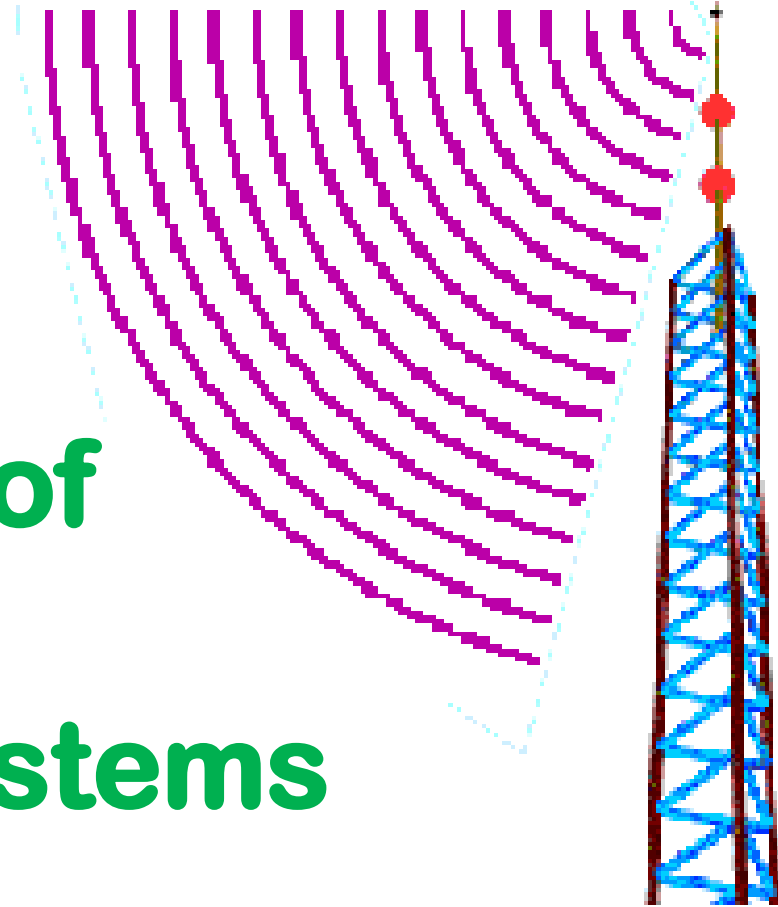


Module-1

Fundamentals of Electronics Communication Systems



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Information, Message, Signal

Information

Communication of knowledge, such as facts, data or opinions



Message

Physical information as produced by source

People, Machines, Measuring instruments etc.



Analog Message

Physical time-variable quantity usually in smooth and continuous form

Information, Message, Signal

Digital Message

Ordered sequence of symbols selected from finite set of elements



Signal

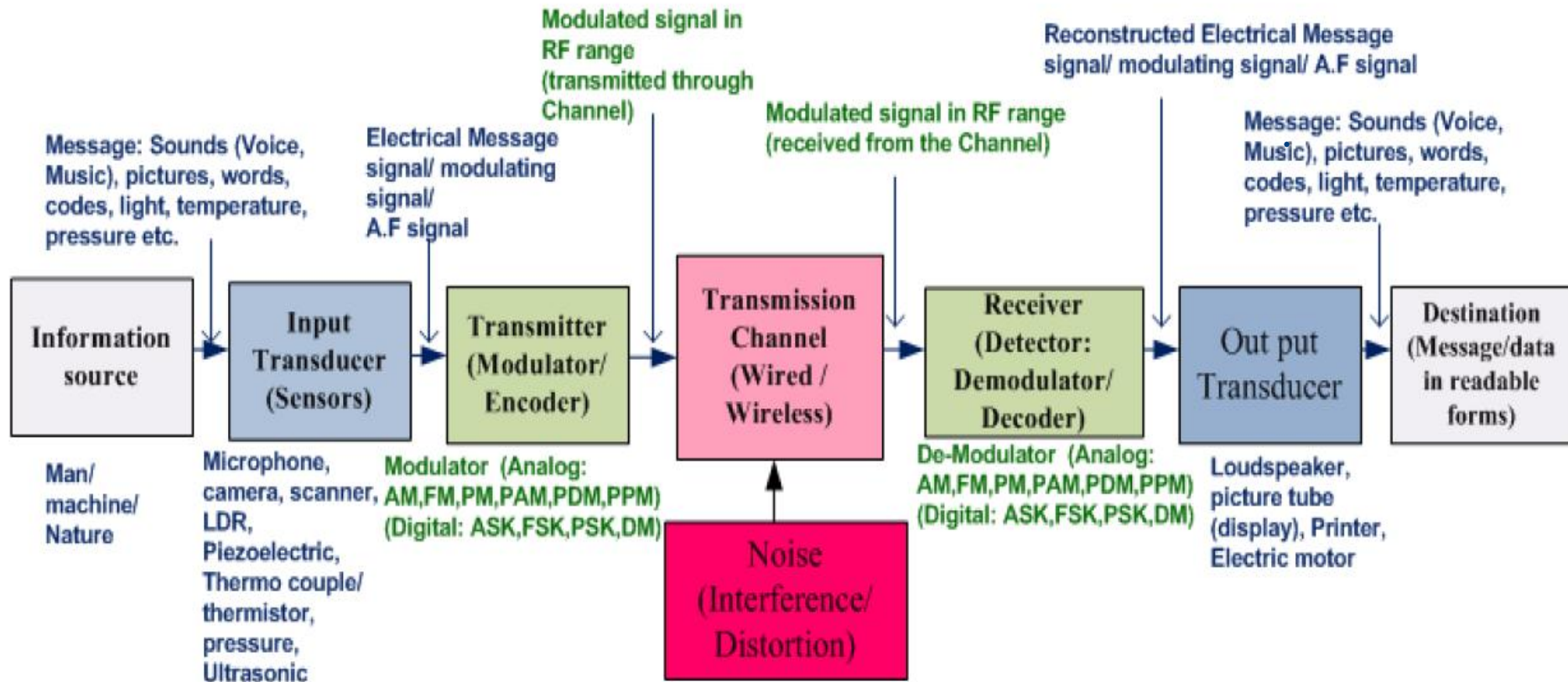
Physical representation of the information



Electrical signal

Voltage or current representing the message

Elements of a Communication System

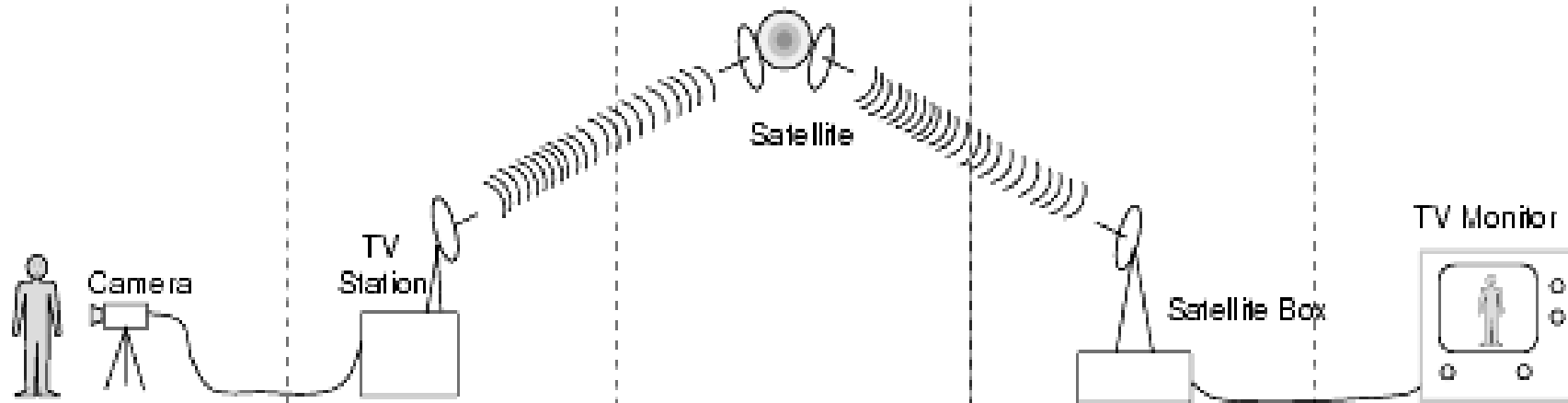


Example-1

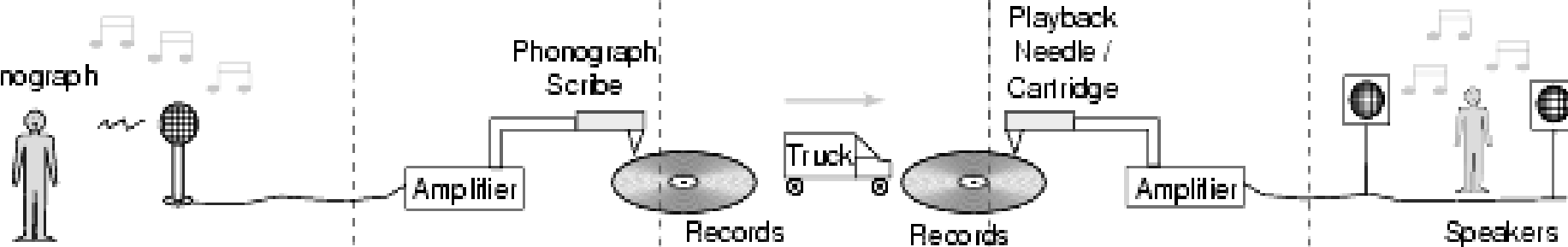
Generic
Communication
System



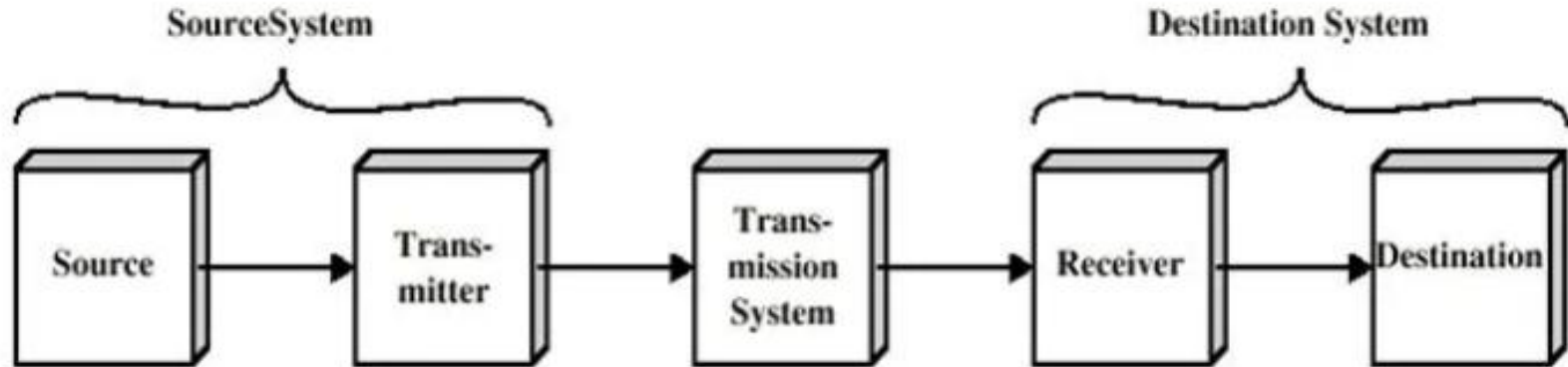
DDS
Satellite TV
System



Phonograph



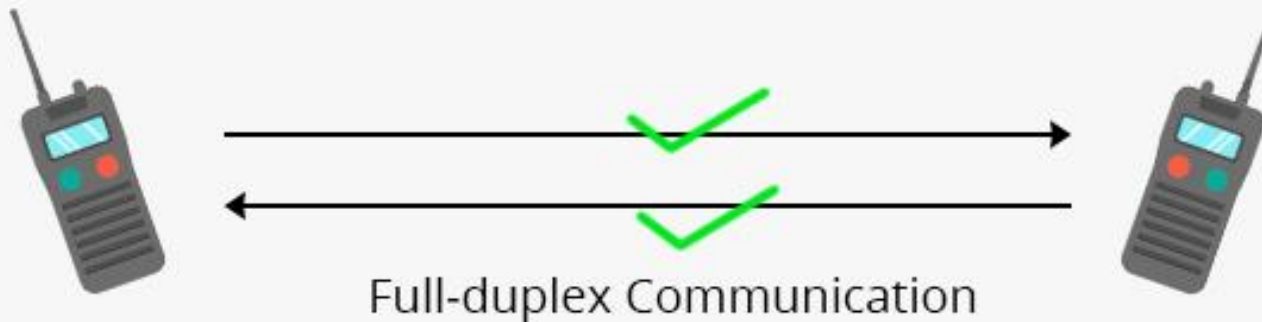
Example-2



(a) General block diagram



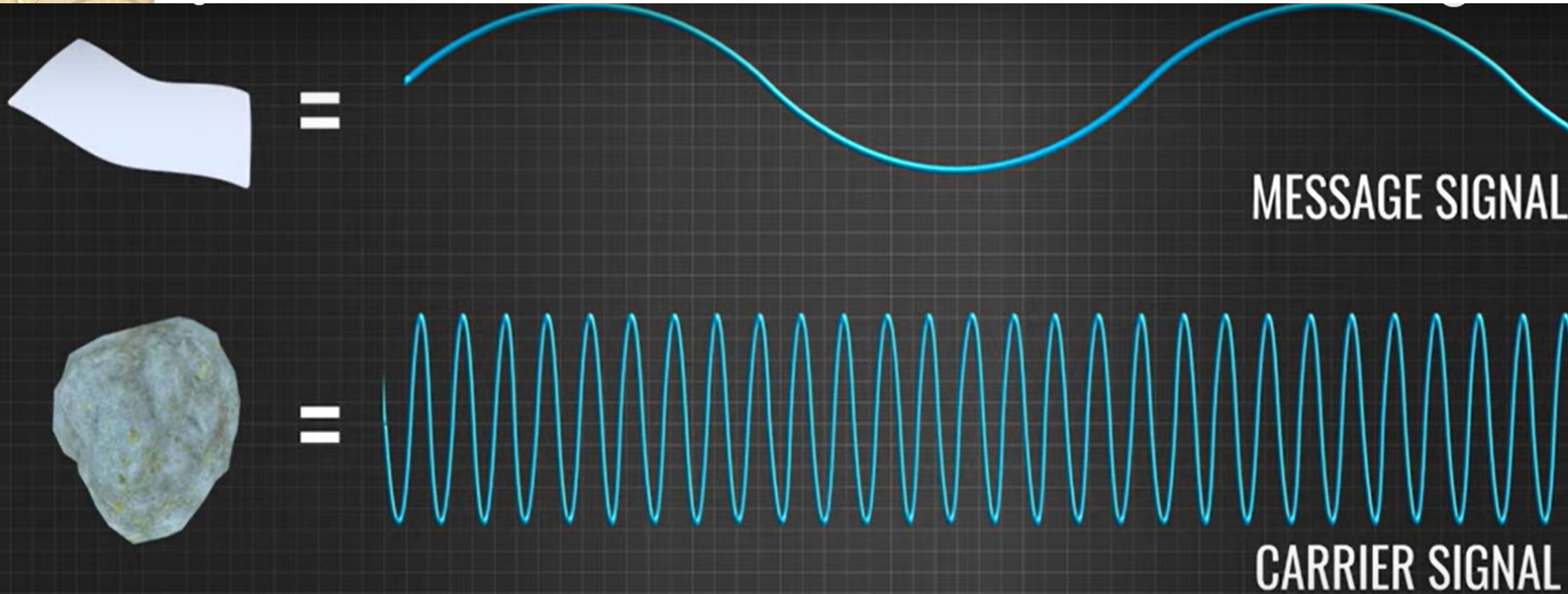
Transmission Modes in Communication Systems



MODULATION

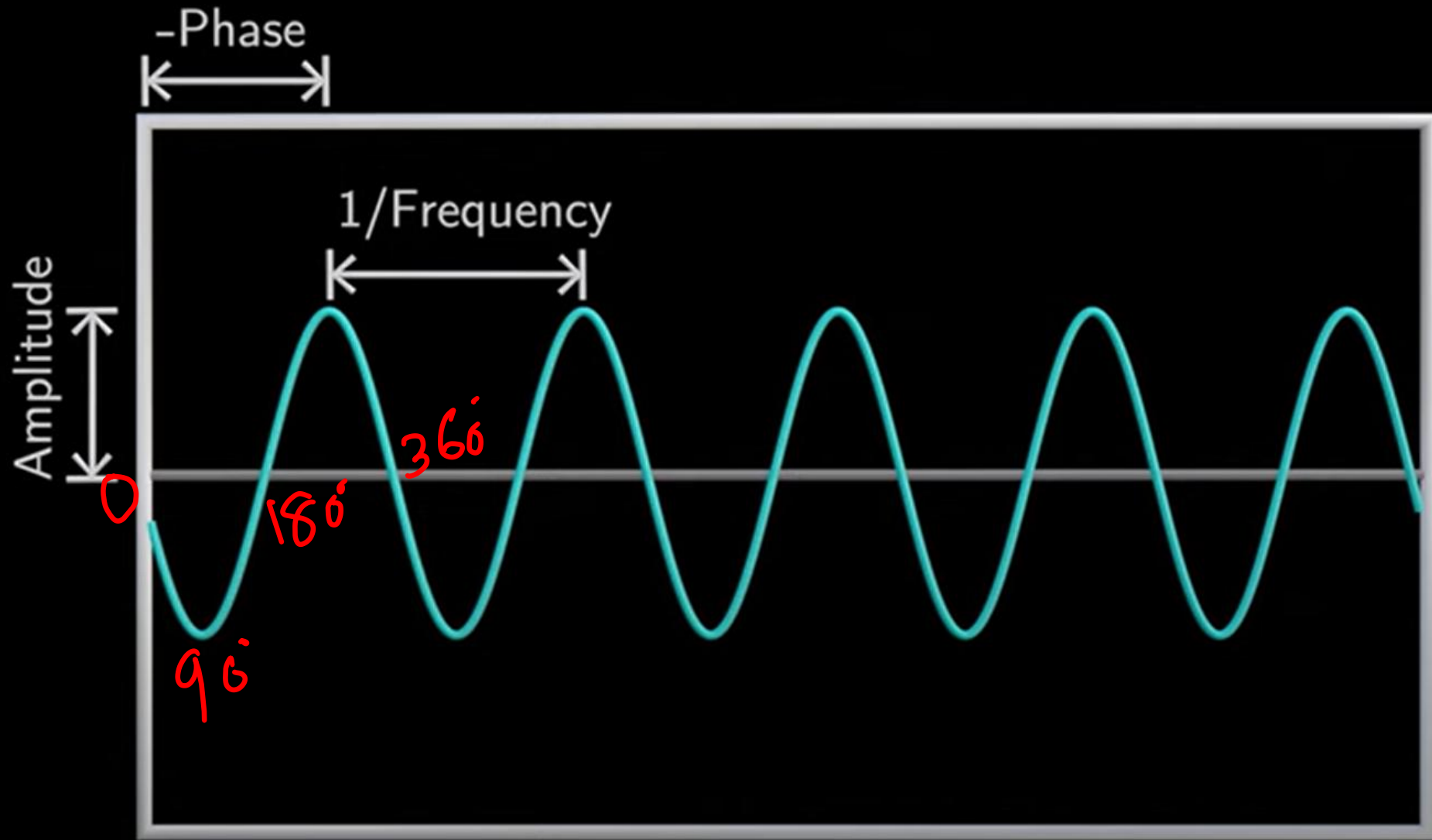
- Example (paper and throw)

MODULATION Components



AMPLITUDE FREQUENCY PHASE

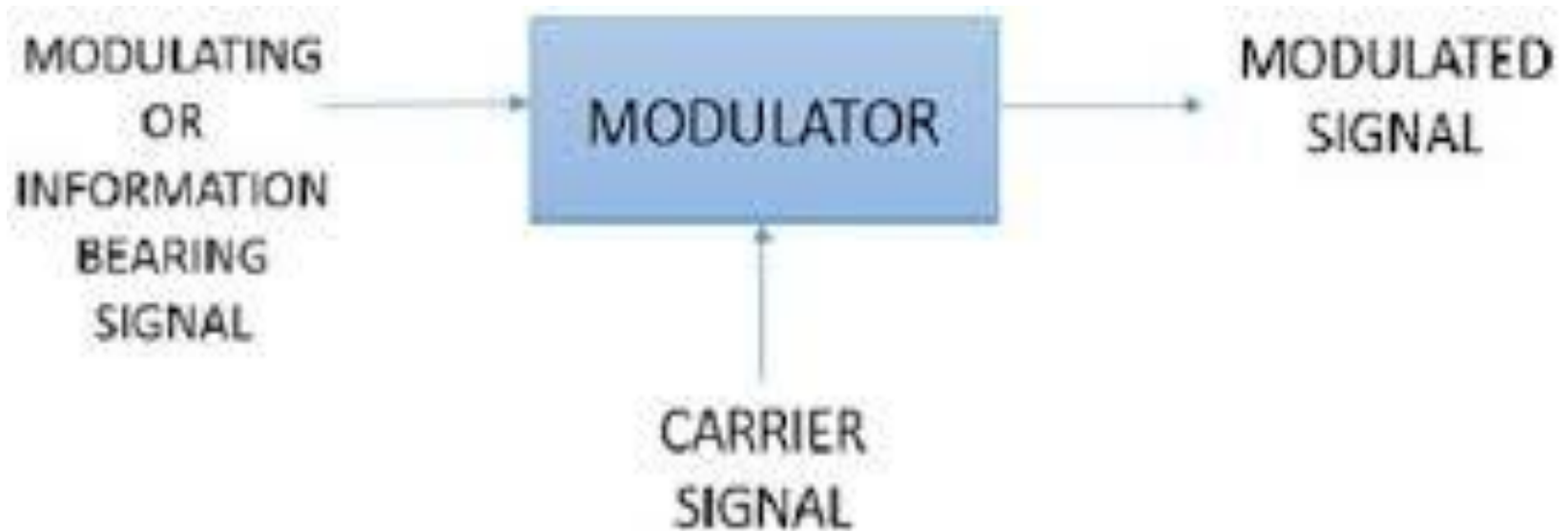
MODULATION Components



Type of Modulating Signal



Process of Modulation

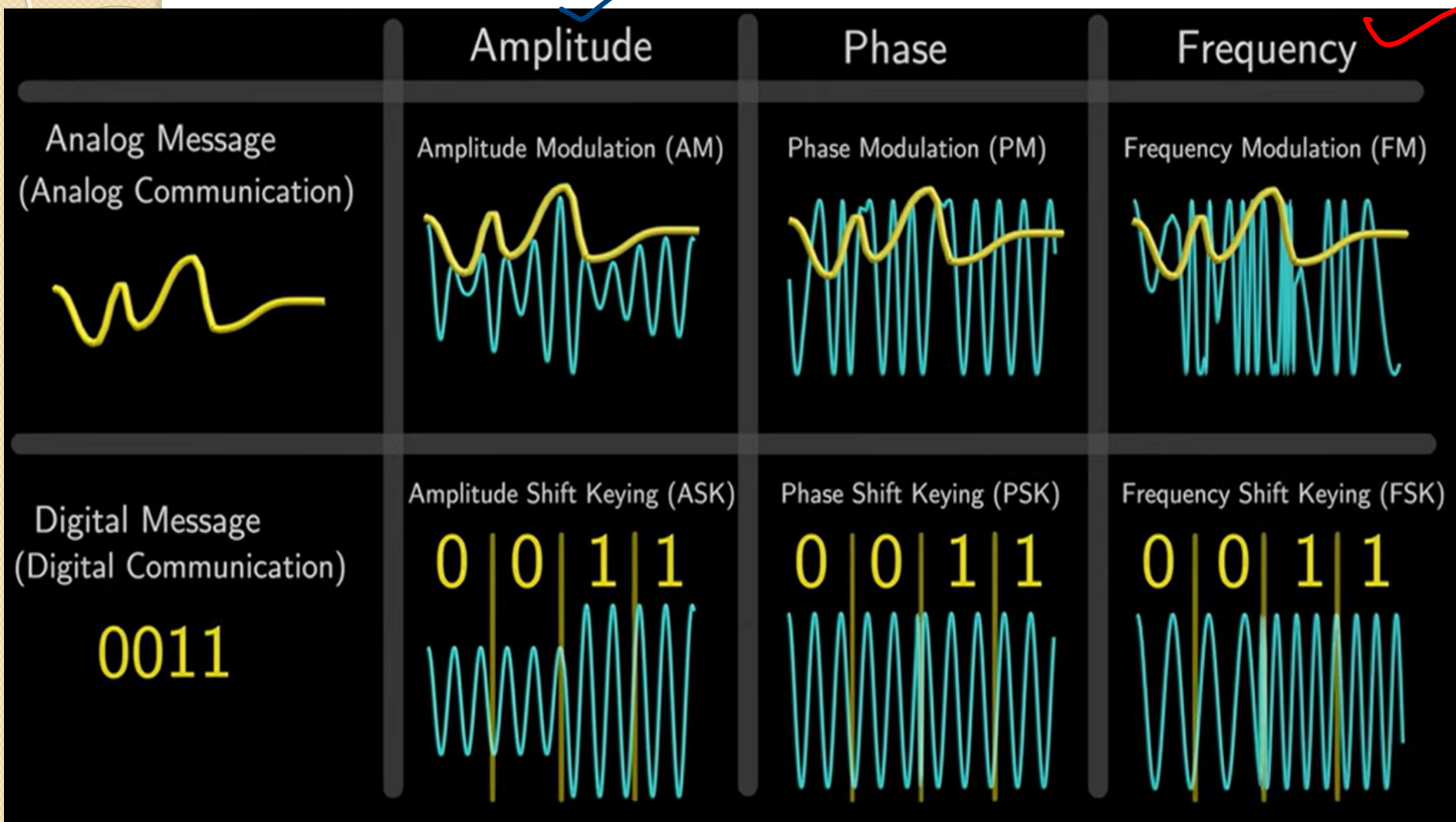


MODULATION

→ Amplitude
→ Frequency
→ Phase

- A Modulation is the process by which some characteristics of a carrier wave is varied in accordance with the instantaneous value of a modulating signal or message signal.
- The carrier wave is changed in accordance with the intensity (i.e amplitude) of the message signal.
- The resultant wave is called modulated wave or radio wave.

Types of Modulation



MODULATION-NEED

I. Reduction in Antenna Height $(\lambda/2)$

$$\text{wavelength} = \frac{\text{Velocity}}{\text{frequency}} = \frac{3 \times 10^8}{\text{frequency in (Hz)}} \text{ meter}$$

Take frequency – 10 KHz and calculate antenna height

$$= \lambda = 30 \text{ km}$$

Take frequency – 100 MHz and calculate antenna height

$$= \lambda = 3 \text{ m} \rightarrow \lambda/2 = \underline{1.5 \text{ m}}$$

MODULATION-NEED

2. Effective Power Radiated by Antenna

- Power radiated is directly proportional to $\left(\frac{l}{\lambda^2}\right)$
- The energy of wave depends on frequency
- Higher the frequency more is the power radiated
- As the audio signal frequencies (20 Hz to 20 KHz) are small they cannot be transmitted over large distances if radiated directly into space.

MODULATION-NEED

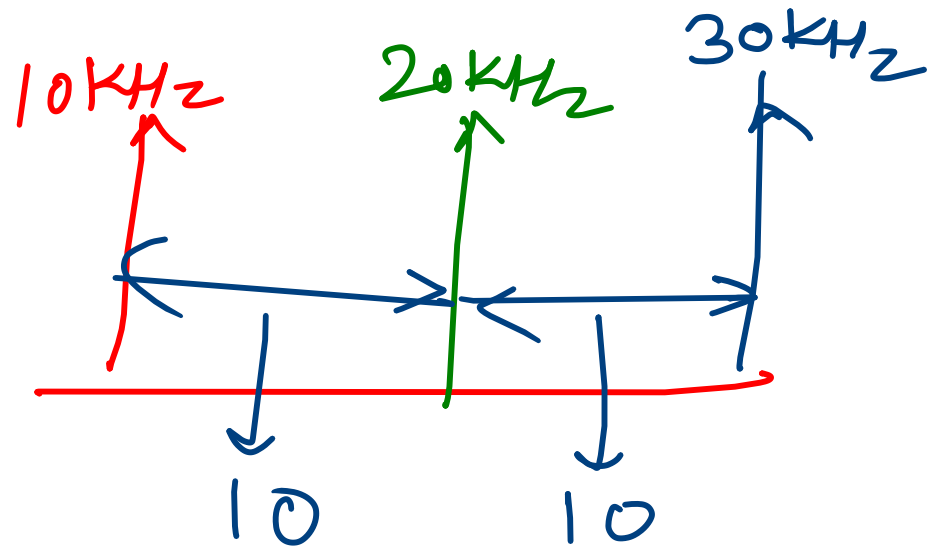
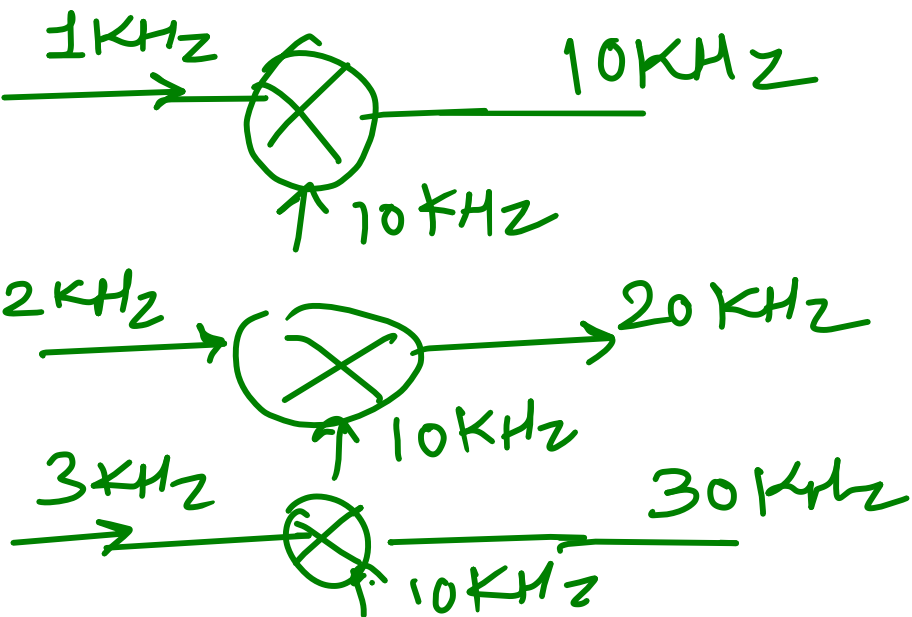
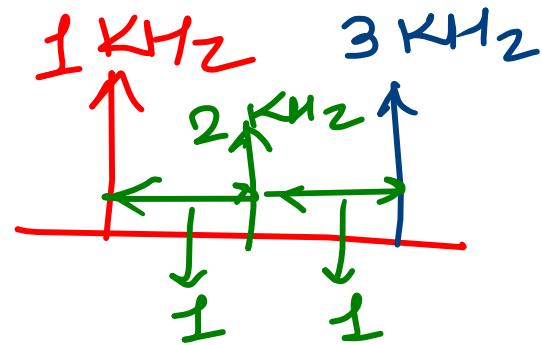
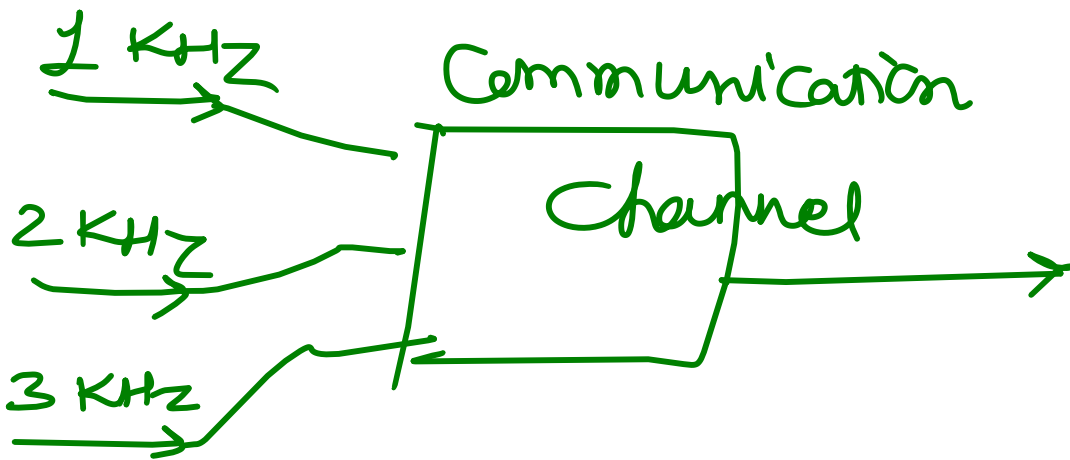
3. Wireless Communication is possible

- Modulation removes necessity of using wires
- One desirable feature of radio transmission is that it should be carried without wires i.e. radiated into space.
- However, efficient radiation of electrical energy is possible at high frequencies (> 20 kHz).

MODULATION-NEED

4. Multiplexing

- If more than one signal uses a single channel then modulation may be used to translate different signals to different spectral location, thus enabling the receiver to select the desired signal.
- Application of multiplexing includes data telemetry, broadcasting and long distance telephones.
- Take signal $f_1=1\text{KHz}$, $f_2=2\text{KHz}$ and $f_3=2\text{KHz}$
- Modulate each signal with carrier of 10KHz
- Resultant signal is placed apart in spectrum

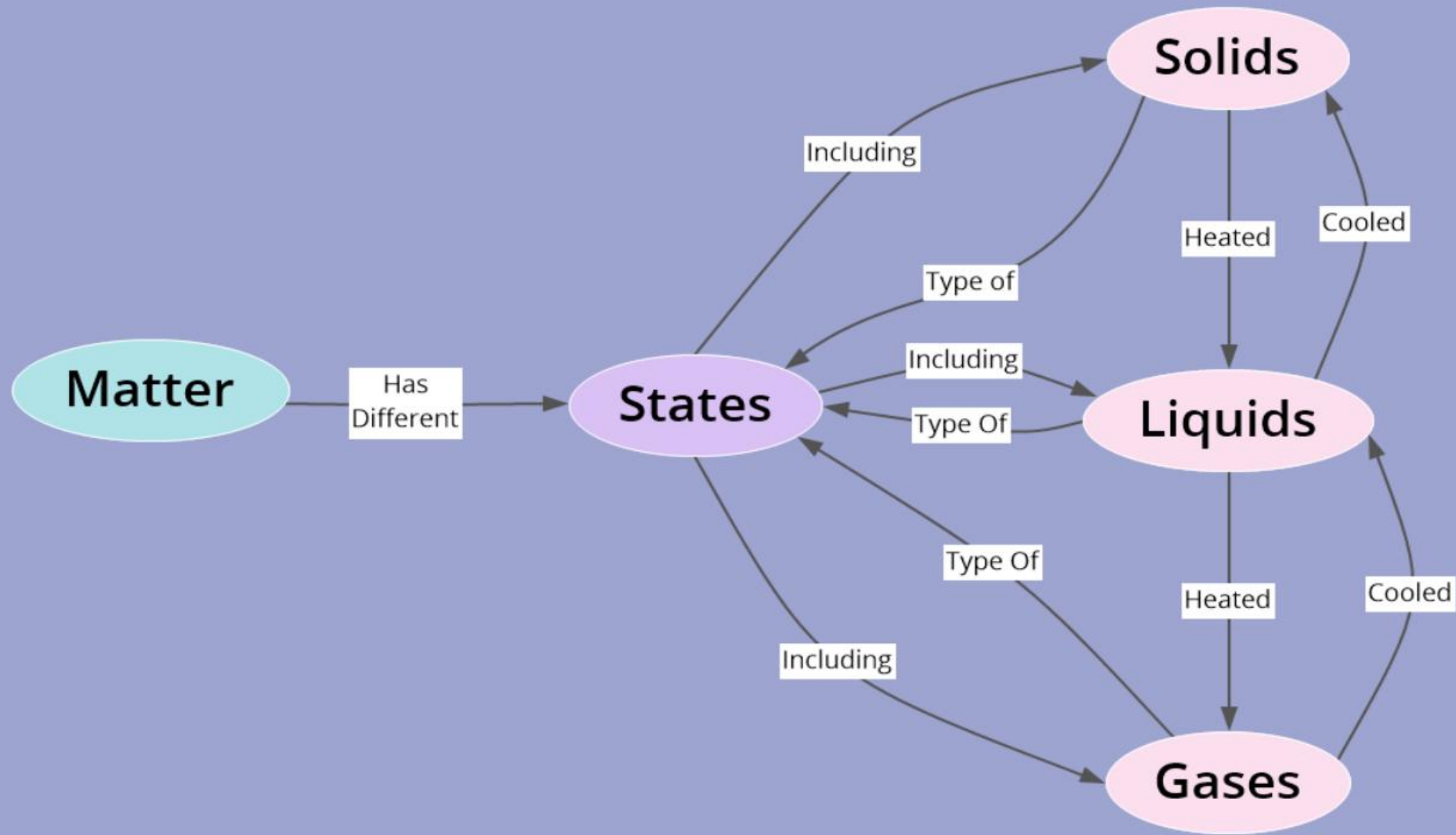


MODULATION-NEED

5. To overcome equipment limitations (translate signal to fix frequency bands)

- If the equipment is elaborate and complex. It is operated in some fixed frequency range,
- thus translating the frequency range of the processing signal corresponding to this fixed range of equipment.
- Modulation can be used to accomplish this frequency translation.

Sample Example of Concept Map



Sample Example of Concept Map

TotalTaxi.at - your airport taxi service in Vienna

